INCH-POUND

MIL-C-70770 (AR)
AMENDMENT 2
17 January 1995
SUPERSEDING
AMENDMENT 1
13 April 1993

MILITARY SPECIFICATION

CONTAINERS, SHIPPING AND STORAGE:
PA 137, PA 138, AND PA 141, FOR
THE M139 MINE DISPENSER SYSTEM

This amendment forms a part of MIL-C-70770(AR), dated 4 November 1992, and is approved for use by the U.S. Army Armaments Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

* Add new paragraph 3.2.1 as follows:

"3.2.1 Cushion fabrication. Cushions may be fabricated from plank material joined by a suitable adhesive to provide 90 percent adhesion between surfaces being joined. The adhesive shall establish and maintain a permanent joint when cushions, installed in loaded shipping containers, are subject to the mechanical environments specified herein."

* 3.3.1.3: Delete in its entirety and substitute the following:

"3.3.1.3 Drop tests. The PA138 and PA141 containers shall be subjected to the edgewise drop test and the PA137 container shall be subjected to the free fall drop test as specified in 4.5.3, followed by a Pressure Retention test as specified in 3.3.1.1."

AMSC N/A FSC 1095

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* 3.3.1.4: Delete in its entirety and substitute the following:

"3.3.1.4 Cornerwise and 7 foot drop. The PA138 and PA141 containers shall be subjected to the Cornerwise drop test and the PA137 shall be subjected to the seven (7) foot drop test as specified in 4.5.4, followed by a Pressure Retention test as specified in 3.3.1.1."

* Add new paragraphs 3.3.3, 3.3.4 and 3.3.5 as follows:

"3.3.3 Laminate strength. Fiberglass reinforced plastic used in the container shall meet the strength requirements of Drawing 12564879 when tested as specified in 4.5.9.

3.3.4 Laminate hardness. Fiberglass reinforced plastic used in the container shall meet the Barcol hardness requirements of Drawing 12564879 when tested as specified in 4.5.10.

3.3.5 Laminate weight. Fiberglass reinforced plastic used in the container shall meet the glass content requirements of Drawing 12564879 when tested as specified in 4.5.11."

* Add new paragraphs 3.4.1, 3.4.2 and 3.4.3 as follows:

"3.4.1 Glass kit procedure. Fabrication of container base and cover shell configuration shall be documented on a glass kit schedule for primary laminates as defined on Drawing 12564879. As a minimum, the glass kit shall define, for each layer, the sequence of application, material description, location, orientation, size, quantity, weight, and overlap of converging layers. The glass kit shall be prepared for each container cover and base shell and documented to record part number, approved revision(s) and date(s). The glass kit schedule and any changes thereto, shall be submitted in writing to the contracting officer or the procuring agency for approval prior to its use in shell fabrication for first article samples and production units. A traveler record shall be maintained which is integral or supplemental to the glass kit. Follow-on production programs with the same manufacture and previously approved glass kit schedule shall not require approval again provided the contractor informs the contracting officer or the procuring agency in writing.

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PAGE 4 (Cont'd)

3.4.2 Lamination process. Departures from the fabrication sequence specified in Drawing 12564879 require approval in writing from the contracting officer. It shall be the responsibility of the contractor to assure that approved glass kit schedules are being followed by qualified personnel in the fabrication of container shells and installation of interior dunnage. At each in-process and final inspection level, traveler record shall be kept for each shell or cover processed through secondary lamination which records the date and personnel who performed each process or inspection operation.

3.4.3 Repair procedure. When necessary and not prohibited by contract, the contractor shall prepare repair procedure for laminates detailing applicable factors of 3.4.1. The procedure shall also include preparation of surface to be repaired, finish after repair, and inspection to ensure removal of defects. Repair procedures shall apply only to defects greater than allowed in Drawing 12564879. Approval in writing by the Government or procuring activity is required prior to use of repair procedures."

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* TABLE I:

SHEET OF Block: Delete "SHEET 1 of 1" and substitute "SHEET 1 of 3".

Add new sheet 2 and 3 as included pages 6A and 6B.

Shipping and Storage Container: PA137 (12564839), Examination for defects: Delete "4.4.2.14" and substitute "4.4.2.17" under Inspection Method Reference.

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* 4.4.2.1: Add new Major 105 as follows:

"105 Pressure retention test failure ...100% ...3.3.1.1 ...Tester, 4.5.1".

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* 4.4.2.2: Add new Major 105 as follows:

"105 Pressure retention test failure ...100% ...3.3.1.1 ...Tester, 4.5.1".

Minor 201, Inspection Method Reference: Delete "Visual/Gage" and substitute "Visual".

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Shipping and Storage Container: Air System	SHEET 2 OF 3	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Base Assembly, Container: PA137 (12564812) Glass kit not approved or shell schedules incomplete Examination for defects	2, 0-1	3.4.1 3.2	Certification 4.4.2.16
	Base Assembly, Container: PA137 (12564770) Glass kit not approved or shell schedules incomplete Examination for defects	2, 0-1	3.4.1 3.2	Certification 4.4.2.6

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Shipping and Storage Container: Air System	SHEET 3 OF 3	DRAWING NUMBER See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cover Assembly, Container: PA138 and PA141 (12564799)	2, 0-1		
	Glass kit not approved or shell schedules incomplete		3.4.1	Certification
	Examination for defects		3.2	4.4.2.12 and 4.4.2.13
	Base Assembly, Container: PA138 and PA141 (12564800)	2, 0-1		
	Glass kit not approved or shell schedules incomplete		3.4.1	Certification
	Examination for defects		3.2	4.4.2.14 and 4.4.2.15

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Minor 202, Inspection Method Reference: Delete "Visual" and substitute "Visual/Gage".

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* 4.4.2.4, Add the following new Majors:

"105	Glass kit documentation	Level II	3.4.1	Certification
106	Laminate hardness	Level II	3.2	Gage
107	Flange thickness	Level II	3.2	Gage
108	Thickness of groove	Level II	3.2	Gage
109	Slot location and dimension	Level II	3.2	Gage
110	Location of handle holes	Level II	3.2	Gage
111	Location of latch holes	Level II	3.2	Gage"

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* 4.4.2.5, Add the following new Majors:

"105	Glass kit documentation	Level II	3.4.1	Certification
106	Laminate hardness	Level II	3.2	Gage
107	Flange thickness	Level II	3.2	Gage
108	Position of groove	Level II	3.2	Gage
109	Slot location and dimension	Level II	3.2	Gage"

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* 4.4.2.6, Delete Major 101 and add the following new Majors:

"103	Bonding of gasket to shell	Level II	3.2	Visual
104	Orientation of arrow to shell stacking feature	Level II	3.2	Visual"

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* 4.4.2.10, Add the following new Majors:

"105	Glass kit documentation	Level II	3.4.1	Certification
106	Laminate hardness	Level II	3.2	Gage
107	Flange thickness	Level II	3.2	Gage
108	Position of groove	Level II	3.2	Gage
109	Slot location and dimension	Level II	3.2	Gage"

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* 4.4.2.11, Add the following new Majors:

"105	Glass kit documentation	Level II	3.4.1	Certification
106	Laminate hardness	Level II	3.2	Gage
107	Flange thickness	Level II	3.2	Gage
108	Position of groove	Level II	3.2	Gage
109	Slot location and dimension	Level II	3.2	Gage
110	Location of handle holes	Level II	3.2	Gage
111	Location of latch holes	Level II	3.2	Gage"

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* 4.4.2.12, Delete Majors 101 and 102. Add the following new Majors:

"103	Bonding of gasket to shell	Level II	3.2	Visual
104	Orientation of arrows to shell stacking features	Level II	3.2	Visual"

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* 4.4.2.13, Delete Majors 101 and 102. Add the following new Majors:

"103	Bonding of gasket to shell	Level II	3.2	Visual
104	Orientation of arrow to shell stacking features	Level II	3.2	Visual"

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* 4.4.2.14, Delete Majors 101, 102 and 103.

Add new Major 106 as follows:

"106	Glass kit documentation	Level II	3.4.1	Certification"

Add the following new Minors:

"203	Handles missing	Level IV	3.2	Visual
204	Latches missing	Level IV	3.2	Visual"

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* 4.4.2.15, Delete Majors 101, 102 and 103.

Add new Major 106 as follows:

"106	Glass kit documentation	Level II	3.4.1	Certification"

Add the following new Minors:

"203	Handles missing	Level IV	3.2	Visual
204	Latches missing	Level IV	3.2	Visual"

Minor 201: Delete in its entirety and substitute the following:

"201	Location of pressure fitting (short lock end)	Level II	3.2	Visual"

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* 4.4.2.16: Delete Majors 101, 102 and 103.

Add new Major 106 as follows:

"106	Glass kit documentation	Level II	3.4.1	Certification"

Add the following new Minors:

"203	Handles missing	Level IV	3.2	Visual
204	Latches missing	Level IV	3.2	Visual"

Minor 201, Delete in its entirety and substitute the following:

"201	Location of pressure fitting (Short lock end)	Level II	3.2	Visual"

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* 4.4.2.17, Add new Major 105 as follows:

"105	Pressure retention test failure	100%	3.3.1.1	Tester, 4.5.1"

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* 4.4.2.18: Delete in its entirety.

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* Add new paragraph 4.4.2.22 through 4.4.2.27 as included.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70770 (AR)
AMENDMENT 2

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Cushion, Strut	1 OF 1	12564808
			NEXT HIGHER ASSEMBLY: 12564800

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Material	Each lot	3.1	Certification
Minor 201	Poor workmanship (general)	Level IV	3.5.1	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70770 (AR)
AMENDMENT 2

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Cushion, Base	1 OF 1	12564809
			NEXT HIGHER ASSEMBLY: 12564800

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Material	Each lot	3.1	Certification
Minor 201	Poor workmanship (general)	Level IV	3.5.1	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70770 (AR)
AMENDMENT 2

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Cushion Assembly	1 OF 1	12564825
			NEXT HIGHER ASSEMBLY: 12564800

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material improper	Each lot	3.1	Certification
102	Bonding of synthetic rubber to cushion assembly improper	Level II	3.2	Visual
Minor				
201	Poor workmanship (general)	Level IV	3.5.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70770 (AR)
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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Cushion Assembly	1 OF 1	12564867
			NEXT HIGHER ASSEMBLY 12564800

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material improper	Each lot	3.1	Certification
102	Bonding of synthetic rubber to cushion assembly improper	Level II	3.2	Visual
Minor				
201	Poor workmanship (general)	Level IV	3.5.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70770 (AR)
AMENDMENT 2

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Cushion Assembly	1 OF 1	12564868
			NEXT HIGHER ASSEMBLY: 12564800

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material improper	Each lot	3.1	Certification
102	Bonding of synthetic rubber to cushion assembly improper	Level II	3.2	Visual
Minor				
201	Poor workmanship (general)	Level IV	3.5.1	Visual

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CLASSIFICATION OF CHARACTERISTICS

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Cushion	1 OF 1	12564869
			NEXT HIGHER ASSEMBLY: 12564800

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material improper	Each lot	3.1	Certification
102	Bonding of synthetic rubber to cushion assembly improper	Level II	3.2	Visual
Minor				
201	Poor workmanship (general)	Level IV	3.5.1	Visual

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* 4.4.3.1, Delete in its entirety and substitute the following:

"4.4.3.1 Structural workmanship testing. The container structure shall be checked for workmanship in accordance with ASTM-D-2563, Level IV as specified on Drawing 12564879."

* 4.5.1:

Line 8: Delete "Each container opening ...retention test."

Line 11: Delete "65 ± 5" and substitute "100 + 20".

* 4.5.3, Delete in its entirety and substitute the following:

"4.5.3 Edgewise drop and 3' free fall drop. The PA138 and PA141 containers shall be subjected to the Edgewise drop test No. 3 while the PA137 shall be subjected to the free fall drop test No. 2 of ARDEC Drawing 8837375 followed by a pressure retention test."

* 4.5.4, Delete in its entirety and substitute the following:

"4.5.4 Cornerwise and seven (7) foot drop. The PA138 and PA141 containers shall be subjected to the cornerwise drop test No. 4 while the PA137 shall be subjected to the seven foot drop test No. 13 of ARDEC Drawing 8837375 followed by a pressure retention test."

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4.5.7, Delete in its entirety and substitute the following:

"4.5.7 Destructive test prior to production. Prior to production and as part of first article, workmanship samples shall be taken and examined for compliance to 3.5.1 and 3.5.2 and dimensionally for thickness. Each destructive test as specified in Drawing 12564879 shall first be performed on the base shell and then on the cover shell. All samples for destructive testing shall be taken along the 0° axis of the triaxial cloth. Sections shall also be taken at bracket secondary lamination as specified on the applicable drawing. Workmanship specimens shall be labeled by part number and section description. The specimens shall be retained as a standard for later use in maintaining good workmanship during production. If any of the specimens fail to meet the requirements listed in Drawing 12564879, the entire shell lot shall be rejected."

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* Add new paragraphs 4.5.8, 4.5.9, 4.5.10, 4.5.11:

"4.5.8 Laminate strength tests. Samples cut from the first article base and cover shall be tested for compression strength, tensile strength, and flexural strength according to the test methods specified in Drawing 12564879. Five specimens shall be taken from each cover and base to do each test. The average of the measured values shall be compared to the minimum physical property value listed in Drawing 12564879. If any of the average values fails to meet the requirement, the container lot shall be rejected.

4.5.9 Laminate hardness tests. The container laminate shall be tested for Barcol hardness according to the test method in Drawing 12564879 to verify requirement of 3.3.4.

4.5.10 Latch mechanism. The latch assemblies on the container shall require no more than 30 pounds force to open and close. The force shall be applied and measured within one inch of each latch lever end in a direction perpendicular to the latch lever. The latch nuts shall be torqued to 100 + 20 inch pounds in the closed position prior to testing. The container shall be tested at ambient temperature and pressure.

4.5.11 Laminate weight test. The container laminate shall have a total glass fiber and triaxial content of 42-53% by weight when tested in accordance with ASTM D2584."

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